

Working Title

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Abstract—Mixed-connectivity traffic allows for changes in responses to sudden road capacity drops from car crashes. Conventional human-driven vehicles don't have the same information about road segment travel times as Connected Vehicles (CVs) do. This study looks at the impact of varying CV penetration rates on road capacity and average delay. Using the Simulation of Urban Mobility (SUMO), a mixed-autonomy fleet is evaluated on the I405 highway and the local road network in Los Angeles. The static human-driven navigation is contrasted with dynamic A* routing in CVs. Average trip delay counterintuitively increases due to CVs taking longer local detours, but overall network efficiency improved via a reduction in average travel time and waiting time with an increase in total completed trips.

Index Terms—SUMO, TraCI, Rerouting, Connected Vehicles

I. INTRODUCTION

The transition from human-driven vehicles to mixed-autonomy involving connected and autonomous vehicles represents a shift in traffic flow dynamics. Testing traffic control strategies in real-world environments is cost-prohibitive and dangerous, so simulating traffic mobility is a great tool for research. The Simulation of Urban MObility (SUMO) provides an environment that allows for high quality testing of traffic scenarios in a simulated environment [1] [15].

A challenge in traffic control is the problem of navigational logic during sudden capacity drops, such as a lane closure from a traffic accident. Conventional human-driven vehicles tend to navigate using shortest-path algorithms. Modeling human-driven vehicles as non-communicative vehicles assumes that the path is calculated once at the beginning of the trip and being unaware of congestion that builds up on a main highway route. Static routing becomes inadequate for events like traffic accidents.

Connected Vehicles can use communication infrastructure to understand a greater picture of the traffic scenario. If a connected vehicle realizes there is a lot of congestion ahead, it can find a traditionally slower route using the local traffic network that has become more efficient during the increased congestion. Varying the number of connected vehicles that can find quicker routes may lead to wait time improvements for both the connected vehicles and the human drivers that don't consider switching routes [8].

This study investigates the macroscopic impact of varying connected vehicle penetration rates on a vehicular network comprising a primary arterial highway and a local secondary system. By procedurally generating a static traffic accident

and comparing greedy static navigation against cooperative dynamic A* routing, this research aims to quantify network resilience. The primary objective is to evaluate overall network efficiency by measuring the average delay of the combined vehicle fleet under adverse capacity constraints.

II. RELATED WORK

A. Microscopic Simulation of Mixed Traffic

Microscopic simulation is widely used to evaluate mixed traffic systems involving connected and autonomous vehicles, as it captures vehicle-level interactions and heterogeneous driving behavior [9]. The Simulation of Urban Mobility (SUMO) platform has been identified as a validated framework for modeling such environments, particularly in studies integrating communication and traffic dynamics [14], [23].

Prior work demonstrates that simulation fidelity depends on appropriate calibration against empirical data, particularly for car-following and lane-changing models [5], [11]. These studies show that microscopic simulators require calibration using real-world traffic data to ensure behavioral realism and accurate travel-time reproduction, as model performance during validation can deviate significantly from calibration results when applied to new traffic conditions [22]. Existing microscopic modeling efforts predominantly focus on car-following, lane-changing, and vehicle interaction dynamics, including extensions of traditional models such as the Intelligent Driver Model (IDM) and Cooperative Adaptive Cruise Control (CACC) frameworks for connected environments [14]. However, existing studies primarily focus on reproducing realistic vehicle dynamics and stability, with comparatively less emphasis on how routing strategies influence network-wide performance under congestion.

B. Penetration Rate Effects on Traffic Performance

The effect of Connected Vehicle penetration rates on traffic flow and congestion has been shown to be non-linear and context dependent [5], [11]. Simulation based studies indicate that increasing automated vehicle penetration can initially reduce roadway capacity due to conservative driving behaviors, such as larger time headways [5].

In contrast, other models demonstrate that higher penetration rates can improve traffic flow when communication enables coordinated or informed behavior among vehicles [11], [16]. These results suggest that system performance

depends not only on the proportion of connected vehicles but also on how information is utilized. Notably, most prior work focuses on mainline traffic performance, with limited consideration of how adaptive routing decisions redistribute traffic across the broader network [4].

C. Connectivity Enabled Routing Behavior

Vehicle connectivity enables real-time access to traffic information beyond local perception, fundamentally altering how vehicles respond to congestion. Vehicle-to-Everything (V2X) communication supports the dissemination of traffic conditions through vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication, enabling awareness of downstream events such as incidents or bottlenecks [23], [26].

Prior studies distinguish between human-driven and connected vehicle interaction mechanisms. Human drivers are modeled as relying on limited visual observation of multiple preceding vehicles, while connected vehicles utilize V2V and V2I communication to access broader traffic information and predicted motion tendencies [16]. These differences of information availability significantly influence traffic flow, with higher connected vehicle penetration improving overall traffic efficiency under idealized conditions [16].

However, most prior studies focus on flow improvement and stability under communication-enabled behavior [23], rather than how this information is used for decentralized route selection. In particular, the impact of self-directed or greedy rerouting behavior, where vehicles independently select alternative paths to minimize travel time, remains insufficiently explored. Such behavior may lead to unintended consequences, including the redistribution of congestion from highways to secondary roads or urban intersections. The Braess paradox explains how this greedy behavior can have a negative impact for everybody [6].

D. Network Level Control and Traffic State Estimation

To address congestion propagation, prior research has explored coordinated traffic control strategies across interconnected networks [4]. These include integrated ramp metering and trajectory control strategies that regulate vehicle flow at merging sections to prevent breakdown of mainline traffic [11].

Hybrid control strategies combining macroscopic and microscopic control have been shown to improve adaptability across different traffic demand levels, but rely on centralized coordination mechanisms and cooperative vehicle behavior. Model-based approaches demonstrate that integrating freeway ramp metering with urban traffic signal control can reduce spill-back effects and improve overall network performance [4]. Similarly, lane-change coordination strategies have been shown to mitigate capacity drops at bottlenecks and enhance traffic flow stability [26].

In parallel, traffic state estimation techniques show that accurate traffic density and flow estimation is possible even at low penetration rates of connected vehicles [13]. A discrete-time macroscopic model based on conservation of vehicles, formulated as a Linear Parameter-Varying system with Kalman

filtering, demonstrates satisfactory density estimation at penetration rates as low as 2%. Ramp flow estimation remains effective at slightly higher penetration rates, while the use of moving averages improves performance under sparse connectivity conditions. [13].

However, these approaches primarily rely on centralized control, estimation, or coordination mechanisms and do not explicitly address how decentralized routing decisions made by connected vehicles influence network-wide congestion patterns under mixed-connectivity conditions.

E. Limitations of Existing Work

Despite extensive research on microscopic traffic simulation, penetration rate effects, connectivity-enabled information sharing, and coordinated traffic control strategies, most existing work focuses on either traffic state estimation or centralized control mechanisms for improving flow stability and capacity [11], [13], [16]. These approaches generally assume cooperative behavior or system-level coordination.

In contrast, comparatively less attention has been given to how decentralized routing decisions made by connected vehicles, based on locally available or communicated traffic information, influence network-wide traffic performance under congestion. In particular, the interaction between mixed-connectivity penetration rates and self-directed route selection remains insufficiently explored in realistic network environments where vehicles independently adapt their routes.

III. METHODOLOGY

A. Simulation Environment

The experimental framework consists of a vehicular network with a primary arterial highway and a secondary local road system. The traffic flow is simulated using the Simulation of Urban MObility (SUMO) [1] platform, and complemented with the Traffic Control Interface (TraCI) to enable real-time manipulation of vehicle behaviors [24]. The highways acts as the default, high-volume route, and the local road network provides an alternative lower-volume/detour routes.

The experiment is based on a real-world section of road as show in Figure 1. It is a three mile section of road in Los Angeles. The main through route is I-405, and it runs in parallel with Sawtelle Boulevard. Our download of the map is six kilometers long by two kilometers wide. The downloaded map layout can be viewed in Figure 2. The six kilometer section allows for enough length to observe upstream routing responses with regards to downstream traffic changes while maintaining reasonable computational resources. It also ensures that there is enough room for newly departing vehicles in SUMO to spawn on time, ensuring that the data analysis works correctly. The two kilometer width acts as a way for vehicles to quickly get off the highway and move parallel to the highway on potentially quicker routes.

The total experiment time is 40 real-world minutes. When the simulation first starts, there are no vehicles populated. We start each simulation with a 10 minute warm up period to get vehicles covering the map and with speed. At the 10 minute

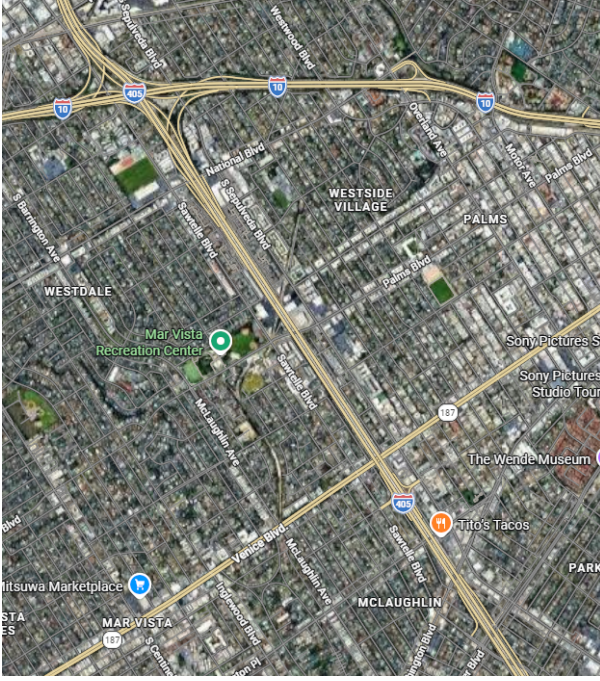


Fig. 1. Satellite view of I405



Fig. 2. SUMO Download of I405

mark, we simulate the crash. Then, we have 30 minutes to observe how the vehicles react to the congestion created by the accident.

B. Justification of Spatial Network Dimensions

Ideally, we would simulate on a very large road network. This would ensure that the simulation is as close to real world outcomes as it can be. It would have a more accurate road network, include more communication between nodes, allow for more paths to consider when navigating around traffic accidents, and have a better simulation of local traffic. The cost comes with the setup time for the simulation and the simulation run time with spawning in vehicles and calculating new routes for vehicles. Notably, for data collection, as the size of the simulation grows, the impact that the simulated accident has on the average data outcomes becomes diluted. So, reducing simulation size reduces noise.

While reviewing similar experiments, we looked at the similar experiments to ours to determine the size of the simulation area we should use. One study used a three mile (4.8 km) stretch of US 75 in Plano, Texas. It had between four and six lanes for the duration of the simulation, and contained seven off- and on- ramps [7]. Another simulation uses a 5km stretch before a vehicle breakdown happens. Yet this experiment has a very narrow surrounding area and focuses on bypassing the breakdown via a overpass junction [25].

Other experiments were more focused on the microscopic focus of car following with CAVs. This requires less space than our focus of modeling rerouting behaviors. This experiment used a highway length of 1150 m, with a justification that modeling microscopic behaviors is impaired when a larger selection of traffic paths are used due to computational constraints [18].

Notably, there are some experiments that use much larger road networks. For example, [27] uses the New York City taxi dataset, which contains over 400,000 intersections. This experiment focuses on a broader example of large-scale llm-agent routing optimization, rather than giving individual vehicles the option to autonomously choose their routes. Using such a large dataset comes at the cost of huge computational complexity which was out of the picture for this study.

Another study focused on a large section of downtown Seattle had a more in-depth simulation of the total environment as well. It included information on pedestrian paths and light rail lines. This experiment did not focus on rerouting during an accident or vehicle breakdown, but it did look at coupled control and traffic dynamics [3]. This Seattle study also focused on developing a framework that could be applied to other cities, which is out of the scope of our study.

C. Simulating Traffic Accident

The stretch of I-405 in our simulation is five lanes wide. To create adequate blockage, we block off three of the five lanes in our simulated accident. We repeatedly add three crashed vehicles to the highway, one in each of the three rightmost lanes. Then, we set the 150 meter section the three lanes in

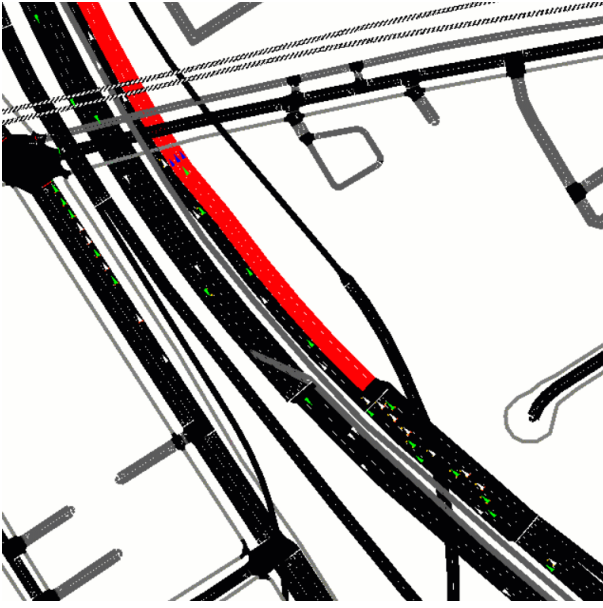


Fig. 3. Accident in SUMO

SUMO to be set to blocked. The remaining two lanes have a lowered speed limit in the area passing the accident.

In Figure 3 you can view a close-up of how the accident is simulated. The closed off sections of roads are in red. The accident/crash vehicles are blue, the human non-connected vehicles are grey, and the connected vehicles are green. The two lanes to the left of the red closed lanes are forced to a lower speed limit for the section adjacent to the red. Figure 3 is a small section of the entire simulation, being a square with about 300m side lengths.

Without the SUMO lane blocking and lowered speed limit, the traffic accident would be a poor simulation of the real world. The two open lanes would remain at highway speeds and there would be a huge backup in the three lanes directly behind the crashed vehicles. Adding the blocked lanes and slower speed allows for the simulation to be much closer to how vehicles respond to real-world accidents.

D. Fleet Composition and Routing

This study models a mixed-autonomy environment with differing levels of Connected Vehicles (CVs) and standard human-driven vehicles. The standard vehicles operate as greedy agents where the navigation relies on static shortest-path algorithms that do not update in live-time, even when there are lane closures due to the simulated traffic accident. Alternatively, the CVs gain traffic information through simulated Vehicle-to-Infrastructure (V2I) and Vehicle-to-Vehicle (V2V) communication to update routing weights and proactively calculate detour routes through the local road network. The communication is grounded in real-world V2X architectures [8] [20]. These routing algorithms will be run with A* search [17]. The connected vehicles recalculate the optimal route every 10 seconds, whereas the unconnected / "human" drivers

maintain the route that was originally planned at the start of its journey.

E. Performance Evaluation

To evaluate the effectiveness of the CVs and the routing algorithms under adverse conditions, a standardized traffic incident is introduced into the network. During each simulation iteration, an accident is procedurally generated at a predefined, static coordinate on the primary highway. This accident systematically obstructs a subset of the lanes, and introduces a bottleneck on the main highway route. The primary variable being measured during the experiment is the average delay of the entire vehicle fleet, combining the results of the connected vehicles and the human-driver vehicles together. This will serve as a qualitative metric for determining overall network efficiency.

Delay is calculated as the actual trip time minus the free flow time.

$$\text{average delay} = \sum_i^N \frac{t_{\text{total}} - t_{\text{free flow}}}{N}, t = \text{time}$$

The free flow time is the theoretically quickest the vehicles route could be completed without any slow downs or stops. It is calculated by assuming the vehicle can travel at the speed limit for every part of the route and seeing how long it would take to cover that distance.

Note that this calculation of free flow time is different from how other real-world calculations are created. For example, the US Federal Highway Administration uses

$$\text{Base Free Flow} = \begin{cases} 40\text{mph}, & \text{speed limit} < 40\text{mph} \\ \text{speed limit} + 7, & 40 \leq \text{speed limit} \leq 50 \\ \text{speed limit} + 5, & \text{speed limit} > 50 \end{cases}$$

but the Free Flow Speed (FFS) calculation uses Base Free Flow Speed as just one parameter in the calculation. It also has adjustment factors for lane width, lateral clearance, median type, and access points [12]. We maintain that our calculation for FFS is suited for the simulation rather than the exact nuances of real world roads. Our calculation for trip delay can be found in our source code. Access to the source code is detailed in Appendix B.

In addition to our custom developed runtime data logs, we took advantage of SUMO's built in statistic generation function `--statistic-output`. This automatically generates XML data to `statistic_output.xml`, and we automatically process the XML data into our output CSV file. This provides us with vehicle trip statistics such as route-Length, speed, waitingTime, departDelay, etc. There is also the category of safety, which contains collisions, emergencyStops, emergencyBraking, and total teleports [1]. This data can be included in output file that is shared in Appendix A and Table I shows a summary of some of these SUMO-generated statistics.

Many of these features are simulation specific, and it allows us to understand if there are deviations from the real world. For example, when we initially set up the accident, the vehicles

Connectivity vs. Various Statistics

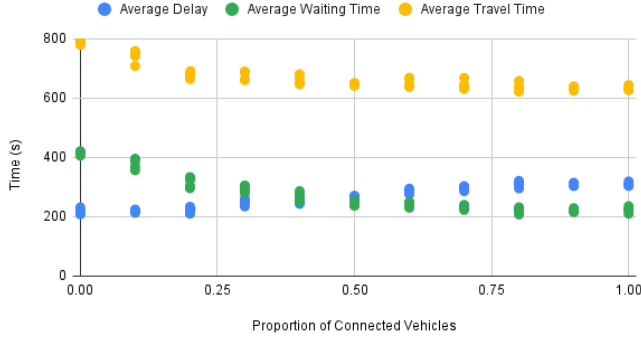


Fig. 4. Proportion of connected vehicles vs. the average delay, travel time, and waiting time of all vehicles in the SUMO traffic flow simulation

Connectivity vs. Completed Trips

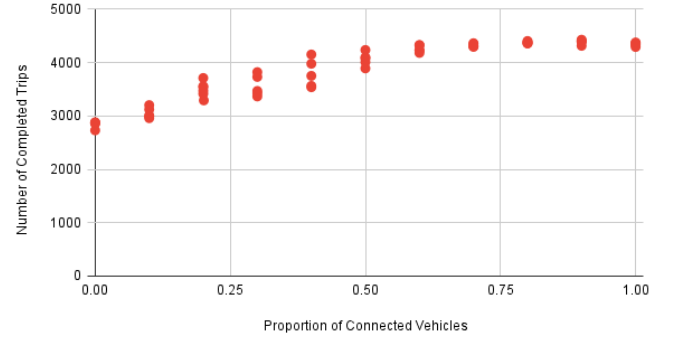


Fig. 5. Proportion of connected vehicles vs. the total number of completed trips in the SUMO traffic flow simulation

would teleport while the experiment was still running. We fixed this by setting `time-to-teleport` value to -1. This disallows vehicles from teleporting after being stationary for some set amount of time.

IV. RESULTS

The data from our experiments are freely available. Access is detailed in Appendix A.

In the data from the experiment, there were four interesting trends. As connectivity increased, the number of trips completed in the time frame increased, the average travel time of all vehicles decreased, the average waiting time (time spent stopped and time spent moving very slowly) of all vehicles decreased, and the average delay (defined above) increased. The data can be seen graphed in Figures 4 and 5.

For the total number of completed trips, while the overall trend was positive, it seems to reach a plateau at around 60% connectivity. The number started around 2,800 and steadily increased until, at 60%, it leveled off at around 4,300. The average waiting time of all vehicles also has a similar plateau, starting at around 410 s and decreasing steadily until around 60% connectivity, where it leveled off at around 220 s. For average travel time, it started around 800 s and showed relatively large improvements when going from 0% connectivity to 20% connectivity. Beyond that, the rate of improvement slows, eventually plateauing at around 640 s. Additionally, the average delay started around 220 s and increased steadily until it plateaued at around 310 s at around 80% connectivity.

V. ANALYSIS

A. Analysis of Results

For three of these statistics (completed trips, average waiting time, and average travel time), the majority of the change happened within around 0-50% connectivity. That is, in our simulation, as connectivity levels near 100%, efficiency returns decrease. The greatest increases in efficiency happened at the lower levels of connectivity, and at higher levels of connectivity, the increases in efficiency were smaller.

This means that (if the model is accurate) with only a small percentage of connected vehicles on the roads, the road systems will see sizable increases in efficiency. However, this may indicate that the simulation is running into an upper limit for these statistics, caused by the limited scope of the simulation. Due to some limiting factor in the simulation (e.g. a bottleneck in the road system), the actual throughput of the system might not be increasing as it should be.

Even if these plateaus are not accurate to reality, the improvements seen in these statistics are still significant. For example, the average travel time decreased from around 800 s to around 640 s over a 6 km long stretch of road. That is a 2 min and 40 s improvement over 6 km, which could amount to up to a 13 min and 20 s decrease in travel time over a 30 km commute.

One trend that was somewhat counterintuitive was the fact that average delay was positively correlated with connectivity (Figure 6), despite the fact that average travel time decreased, average waiting time decreased, and the number of completed trips increased. This indicates that at higher levels of connectivity, vehicles were both delayed more, but also had lower travel and wait times, which makes little intuitive sense.

One possible explanation for this trend is that with more connectivity and better global pathing, cars might be taking more detours onto side roads. Due to the way that free flow time was calculated, cars traveling on side roads would have to deal with more delays that are not factored into free flow time, such as traffic lights, turns, and intersections, thus increasing the difference between actual travel time and free flow time. In addition, a car traveling at 90% of free flow time on a longer path will always have a longer average delay than a car traveling at 90% of free flow time on a shorter path. This means that even very efficient usage of side roads might still result in an increase in average delay. Therefore, because connected vehicles more often take longer paths using more local roads, it makes sense that as the proportion of connected vehicles increase, the average delay would increase as well.

In the end, these explanations are only a few possible

Connectivity vs. Average Delay

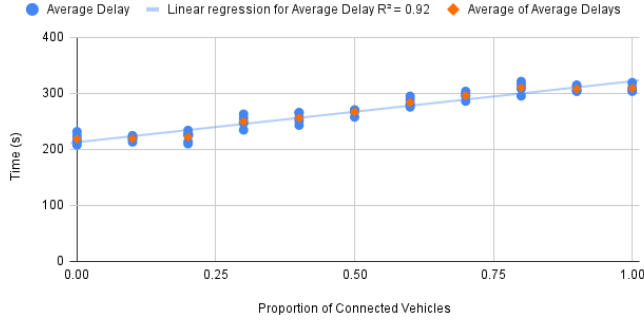


Fig. 6. Proportion of connected vehicles vs. the average delay of all vehicles in the SUMO traffic flow simulation, with linear regression line and averages for each level of connectivity

mitigating factors. In order to come to a more solid conclusion, more testing and experiments have to be done.

The overall conclusion is that increased levels of connectivity generally lead to fuller utilization of the entire road system, which increases the overall capacity and efficiency of the road system. Even at low levels of connectivity, there are significant improvements in efficiency, but it remains to be seen if, in the real world, the improvements will plateau as seen in the experiments. Additionally, more testing and experimentation must be done to verify the true cause of the increase in average delay seen in the experiments.

B. Statistical Significance

TABLE I

ANOVA F-STATISTICS AND P-VALUES FOR VARIOUS MEASUREMENTS FOR A SINGLE FACTOR TEST OVER ALL CONNECTIVITY LEVELS

Statistic Name	F	P-value	F crit	α
Average Delay	107.2731	1.62e-27	2.053901	0.05
Completed Trips	103.9439	3.13e-27	2.053901	0.05
Average Waiting Time	192.5449	6.57e-33	2.053901	0.05
Average Travel Time	81.25003	5.29e-25	2.053901	0.05

In order to prove that these results are statistically significant, single factor Anova tests were performed on all four statistics mentioned in the previous section (Average Delay, Completed Trips, Average Waiting Time, and Average Travel Time) [21]. These tests returned the F-statistics and p-values seen in Table I.

For single factor Anova tests, the F-statistic is the variance between groups divided by the variance within each group. Generally, the higher this number is, the more likely it is that the differences between group means are significant. In other words, the higher the F-statistic is, the more likely it is that connectivity levels had a statistically significant impact on these measurements.

Anova: Single Factor F-Statistics for Various Statistics

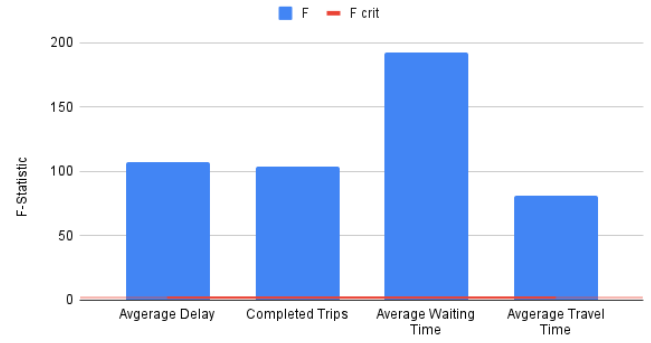


Fig. 7. Anova F-statistics for various measurements for a single factor test over all connectivity levels

The p-value is essentially the probability that the F-statistic occurred by pure chance. For testing statistical significance, the conventional standard of $\alpha = 0.05$ was used. This means that a p-value below 0.05 indicates statistical significance.

The F critical value (F crit) is simply the value of the F-statistic for which the p-value would equal 0.05, so an F value lower than the F critical value indicates no statistical significance, while an F value higher than the F critical value indicates statistical significance.

As can be clearly seen in Figure 7, all of the F-statistics for these four variables were well above the F critical value, indicating that changing the levels of connectivity did have a statistically significant impact on these variables.

VI. LIMITATIONS

As mentioned above in the Analysis section, there are many limitations of the current system that leave many questions to be answered.

A. Size Limitations

There are several aspects of our experiment that could be improved simply by having more time, computing power, and manpower. With more resources, we could have potentially increased the scope of the experiment, running it over a larger area, over a larger time frame, and running it more times.

The road system that is covered is relatively small, only around $12km^2$. By increasing the area of the simulation, it could provide more evidence on whether the plateaus in the simulation are realistic or not. However, as the size of the simulation increases, it becomes more challenging to see the effects of the accident (what we are trying to test) in our resulting data because of the extra noise from traffic that does not navigate near the accident area.

Though, the size of the road system does have an impact on how the simulation is run. Departure delay varies between runs, and it is difficult to understand how exactly this impacts the results from our experiment. Appendix C shows the data of departure delays standardized by total vehicle trip count.

In addition, the simulation was limited to 10 minutes before and 30 minutes after the accident, and the simulation was only run 5 times each for 11 different levels of connectivity. With a larger time frame and more iterations, the data would be more consistent, with less uncertainty and variability.

B. Simulation Inaccuracies

There are other aspects of our simulation that could be improved to make it more accurate to reality, and thus make its results more applicable to real life.

For one, the vehicle flows (the number of incoming vehicles and where they are headed) have been somewhat arbitrarily assigned, and are not completely accurate to reality. For example, in reality, the vehicle flows might start or end within the chosen road network, such as for the road network of an entire city.

Additionally, the behavior of the vehicle models have not been compared to the real life behavior of human driven vehicles and autonomous vehicles. In future experiments, carefully adjusting the attributes of the models to bring them closer to the behavior of their real life counterparts could help immensely in making the simulation more accurate.

In regards to real-world accuracy, one approach that has been taken in other research papers [2] is collecting real world data from a road network and comparing the simulated data to the real data over several key metrics. This would ensure a baseline level of accuracy, from which the levels of connectivity can be varied, and more accurate conclusions can be drawn.

Another limitation that could be addressed in future research is the lack of different types of vehicles and pedestrians in our model. Different types of vehicles may have different levels of access to different roads, as well as different physical properties and different rules. Pedestrians could also change significantly results through their interactions with vehicles. For example, pedestrians taking crosswalks could extend the amount of time that vehicles must stay stopped at intersections, thus increasing the penalty for taking local roads. Especially now that SUMO can simulate pedestrian dynamics in detail, including at intersections, this may be helpful for creating a more realistic simulation [10].

Introducing more types of vehicles also opens the door to introducing multi-modal transportation. With multi-modal transportation, a simulated passenger could take a personal AV to a bus stop, ride the bus for some time, and then walk the remainder of the way. This type of simulation would be particularly applicable when simulating more urban environments. If a future researcher were to try to simulate an entire city's traffic network, they may want to look into simulating multi-modal transportation as well. Concerning CAVs, exciting transportation opportunities such as shared AVs would be something that could be additionally simulated in SUMO and give different routes for additional research [19].

VII. CONCLUSION

As we head into large-scale implementation of Connected AVs, we will have to deal with many levels of mixed traffic.

The main question that this paper addresses is, "How do the levels of connectivity affect the traffic flow when a crisis occurs?" To this end, a simulation was implemented in SUMO using a real-world road system in California. Using SUMO, an accident was simulated on this highway, and multiple simulations were run using different levels of connectivity.

Connectivity improved the overall efficiency and utilization of the road system, but causes a number of cars to have longer delays for the sake of the whole system.

APPENDIX A DATA AVAILABILITY

The dataset generated and analyzed during the simulation is available at https://github.com/qfransen/CSE834_Experiment/blob/main/data.csv

APPENDIX B SOURCE CODE

The SUMO simulation files, TraCI scripts, and routing algorithms used in this study are open-source and available on GitHub. You can access the repository here: https://github.com/qfransen/CSE834_Experiment

The computational experiments presented in this study utilize the open-source microscopic traffic simulation suite, Eclipse SUMO (Simulation of Urban MObility). If you plan on executing the source code and reproducing the simulations provided in the GitHub repository, you must first install the SUMO software package. Furthermore, executing the Python control framework requires configuring the `SUMO_HOME` environment variable on the host machine to allow the scripts to interface with the Traffic Control Interface (TraCI) API. Full installation distributions and official documentation can be accessed via the Eclipse Foundation.

To ensure reproducibility, the repository is modularized into two primary directories:

- **networks/**: Contains all static SUMO-specific XML assets, including the network architecture, traffic flow routes, and configuration files (`.net.xml`, `.rou.xml`, `.sumocfg`).
- **src/**: Houses the core Python execution scripts, agent logic, and data processing utilities. The primary entry point, `launch_experiment.py`, accepts the connectivity penetration rate as a command-line argument and utilizes the Traffic Control Interface (TraCI) to manipulate the SUMO environment. This script programatically governs the 2400-step simulation, dynamically spawns the multi-lane traffic accident on Northbound I-405 at the 10-minute mark (step 600), restricts lane access, and records live vehicle delays.

APPENDIX C AVERAGE DEPART DELAYS

Table II shows the average of total departure delay divided by the total trips started for each level of connectivity. There is a 23.4 s difference between the highest (0.2 connectivity) and lowest (0.7 connectivity) values. A function outputting

these values is non-convex, which makes it challenging to understand the impact this has on our results.

Connectivity Amount	Average Depart Delay / Total Vehicle Count
0.0	153.73
0.1	166.02
0.2	167.05
0.3	163.26
0.4	162.69
0.5	154.83
0.6	145.44
0.7	143.65
0.8	146.47
0.9	150.89
1.0	157.10

TABLE II
AVERAGE DEPART DELAY / TOTAL VEHICLE COUNT

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CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Ali Ali: Supervision, Software, Writing – Review & Editing.
Quinn Fransen: Conceptualization, Methodology, Software, Investigation, Data Curation, Writing - Original Draft. **David Oh:** Conceptualization, Methodology, Software, Data Analysis, Writing – Original Draft.

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